Worksheet 2-2 A
Definition of the Derivative and Derivative Notation #1
AP Calculus Name_____

You must use the definition of the derivative to solve the following problems. Do all work on notebook paper!

1. Given $g(x) = x^2 - 3x + 1$, find the following:
   a.) $g'(x)$
   b.) $g'(-3)$
   c.) $g(-3)$
   d.) Write the equation of the tangent line for $g(x)$ at $x = -3$.

2.) Given $h(x) = 2\sqrt{x}$, find the following
   a.) The average rate of change for h on the interval [0, 4]
   b.) $h'(x)$
   c.) Write the equation of the tangent line for $h(x)$ at $x = 9$

3.) Given $p(x) = \frac{1}{x}$, find the following
   a.) $p'(x)$
   b.) Write the equation of the tangent line for $p(x)$ at $x = 2$

4.) Given $k(x) = 2x - 6x^2$, find the following
   a.) $\lim_{h\to 0} \frac{k(x+h)-k(x)}{h}$
   b.) Explain the meaning of $k'(3)$. Do NOT find $k'(3)$.

Each of the following limits is the definition of a derivative. For each of the following limits, identify $f(x)$.

5.) $\lim_{h\to 0} \frac{((x+h)^2-2(x+h)+1)-(x^2-2x+1)}{h}$

6.) $\lim_{h\to 0} \frac{\sin(x+h)-\sin(x)}{h}$

7.) $\lim_{h\to 0} \frac{e^{(x+h)}-e^x}{h}$

8.) $\lim_{h\to 0} \frac{\arcsin\left(\frac{1}{2}+h\right)-\arcsin\left(\frac{1}{2}\right)}{h}$

9.) $\lim_{h\to 0} \frac{\ln(x+h)-\ln(x)}{h}$

10.) $\lim_{h\to 0} \frac{\sqrt{2(x+h)-3}-\sqrt{2x-3}}{h}$

11.) $\lim_{h\to 0} \frac{\sin\left(\frac{\pi}{3}+h\right)-\sin\left(\frac{\pi}{3}\right)}{h}$

12.) $\lim_{h\to 0} \frac{e^{(3+h)}-e^3}{h}$